

A Step Forward for Medical LLMs in Brazilian Portuguese: Establishing a Benchmark and a Strong Baseline

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Abstract—The application of large language models in healthcare presents unique challenges, particularly in non-English contexts where linguistic and cultural nuances significantly impact model effectiveness. In this work, we introduce a novel benchmark for evaluating medical language models in Brazilian Portuguese, addressing a critical gap in AI assessment for healthcare applications. This benchmark is built upon Brazilian medical aptitude tests spanning 2011-2024, enabling extensive evaluation of both specialist and general large language models. Our findings demonstrate that despite advancements in language model capabilities, significant gaps remain in their ability to reason effectively about medical knowledge in Brazilian Portuguese. This benchmark establishes a proper foundation for evaluating and advancing medical language models in Portuguese, creating a standardized framework to guide development toward more effective, equitable, and culturally appropriate AI systems for healthcare in Brazil.

Index Terms—Machine Learning, Large Language Models, Benchmark, Medicine.

I. INTRODUCTION

The ever-growing progress of artificial intelligence (AI) has influenced several sectors of society, with advances in several areas of computer vision and natural language processing. Among these sectors, healthcare, due to its critical nature of dealing with human lives, has received a great deal of attention, particularly in the analysis of medical images, where convolutional neural networks achieve performance levels comparable to those of experts [1]. Furthermore, AI technologies such as generative chemistry, machine learning, and multiproperty optimization have contributed to the advancement of numerous compounds into clinical trials [2].

Despite significant advancements in AI applications within healthcare, the emergence of generative models has opened a new path for their use. Large Language Models (LLMs) have demonstrated remarkable proficiency in understanding and generating human-like text, leading to their adoption in areas such as clinical decision-making, patient interactions, and medical training. However, evaluating the effectiveness of these models in medical contexts presents unique challenges.

Still, the use of LLMs in such areas should be considered with care, as hallucinations can be particularly dangerous in these scenarios. In the healthcare industry, socio-cultural nuances, such as region-specific diseases, may not be adequately captured by generalist models. As such, several instances of medical-specific models have been proposed in the literature to address these issues [3]–[5].

It is a fact, however, that LLMs often struggle to generalize effectively in less-represented languages, such as Portuguese, due to the limited linguistic diversity in their training dataset [6]. Many multilingual language models are primarily trained on texts in languages like English and Chinese, leading to suboptimal performance when applied to medical terminology and concepts in other languages [7]. This deficiency not only affects their ability to process domain-specific language but also introduces the risk of overlooking culturally relevant nuances, which may affect the model’s utility and fairness in non-English-speaking healthcare contexts. Such a problem that is already relevant in generalist models becomes even more worrisome when considering healthcare.

Another issue is the fact that evaluating LLMs in languages like Portuguese is not well-established due to the absence of standardized benchmarks for assessment. As a result, evaluations often rely on qualitative measures rather than objective, reproducible metrics, making it difficult to evaluate model performance consistently across different languages, especially in such a way that highlights these sociocultural challenges. This lack of clear evaluation frameworks contributes to a significant gap in understanding how well these models perform in non-English medical contexts, limiting their practical application and effectiveness in global healthcare settings.

The linguistic disparity between widely represented languages and Portuguese creates substantial challenges for developing effective medical AI applications. Current resources in Brazilian Portuguese are severely limited, with available datasets suffering from significant shortcomings. One dataset is a translated version of a forum-obtained dataset [8], which lacks clinical validation and contains informal language

patterns inappropriate for medical contexts. The alternative corpus, extracted from 16th-century medical texts, employs archaic terminology and concepts disconnected from contemporary medical practice [9].

The absence of valid Brazilian Portuguese medical datasets not only hinders model development but also affects evaluation capabilities. Without standardized benchmarks specifically designed for Brazilian Portuguese, performance assessment relies primarily on qualitative judgments rather than objective metrics. This evaluation gap makes it difficult to measure model performance consistently across healthcare applications and identify improvements needed for clinical deployment. Given these challenges, this paper introduces the following contributions to advance medical language models for Brazilian Portuguese:

- We introduce a novel benchmark focused on the evaluation of medical language models in Brazilian Portuguese, built using medical aptitude tests.
- We conduct an extensive evaluation of state-of-the-art multilingual and monolingual models, demonstrating that current approaches fail to perform adequately on this benchmark, thus revealing a need for specialized models.

The rest of this work is organized as follows: Section II reviews relevant literature in medical language model evaluation, Section III details our data collection process and benchmarking framework, Section IV presents our experimental findings and analysis, and Section V summarizes our contributions and discusses future directions.

II. RELATED WORKS

Evaluating medical language models has always been a challenge due to the quality and availability of domain-specific data. This issue is particularly relevant for low-resource languages such as Brazilian Portuguese, where the lack of standardized datasets hinders both model development and systematic evaluation.

Several benchmarks have been proposed for medical NLP in high-resource languages like English. MedMCQA [10] and MedQA [11] focus on multiple-choice medical question answering, evaluating a model’s ability to handle clinically relevant knowledge. MMLU [12] includes a medical reasoning category as part of a broader general-knowledge benchmark, providing a comparative assessment of medical understanding. PubMedQA [13] introduces a dataset derived from biomedical literature, testing models on real-world clinical and scientific question answering. Despite these advances, no equivalent benchmark exists for Brazilian Portuguese, limiting the ability to measure and compare model performance systematically in this context.

Although certain models have been proposed for medical NLP in Brazilian Portuguese, they typically rely on qualitative evaluation across distinct tasks, making direct comparison difficult. Clinical-BR-LLaMA [14] is a domain-adapted LLaMA model trained on Brazilian medical texts, but its evaluation lacks a standardized benchmark. Similarly, DrBode [7] presents a medical chatbot designed for Brazilian healthcare

scenarios, yet its assessment is based on task-specific performance rather than a general evaluation framework. The absence of a dedicated benchmark for Brazilian Portuguese medical NLP prevents rigorous model assessment and hinders progress in the field.

This work addresses these gaps by introducing a standardized benchmark, composed of a curated collection of questions from aptitude tests, specifically designed for evaluating medical language models in Brazilian Portuguese, enabling systematic model comparison and guiding future advancements in the field.

III. METHODOLOGY

In this study, we introduce a benchmark and an extensive evaluation of various language models to assess their performance in the medical context by answering questions from a series of Brazilian medical aptitude exams for graduated students, such as the Brazilian National Exam of National Examination for Revalidation of Medical Degrees Issued by a Foreign Higher Education Institution (Revalida) and Residency Entrance Exams from the University of São Paulo (FUVEST), in order to assess their proficiency in Brazilian Portuguese. The primary objective was to identify the language model that provided the most accurate answers to medical questions and demonstrated superior understanding and generation of responses in Brazilian Portuguese.

A. Data collection and cleaning

The first step of the benchmark creation involved collecting question-and-answer data from Revalida and FUVEST. The extraction process involved the usage of regular expressions (regex) to isolate and remove irrelevant content, such as footers and headers. The extracted data from Revalida was organized into ten distinct text files, one for each year of the examination, except for 2017 and 2020, due to the unavailability of data. For the FUVEST Residency Exam, only the general practice exam was used for now. Several issues were identified during this extraction, including misplaced spaces between words and encoding inconsistencies. Despite extensive efforts to clean the data, these challenges persisted.

To address these issues, OpenAI’s GPT4-o mini was employed to rectify the text automatically in order to reassemble words that had been split erroneously and to resolve encoding problems, thereby ensuring the integrity of the dataset. After the initial reconstruction, a manual review was conducted to ensure accuracy. Word differences between the original questions and the reassembled were checked using text comparison tools, and any discrepancies were corrected to maintain the consistency of the dataset.

Besides textual content, certain questions included images, which were manually extracted due to the absence of standardized formatting that would allow for automated extraction. To ensure a more comprehensive understanding of the questions containing images, textual descriptions were created with the assistance of GPT4-o mini. However, it is important to note that these descriptions do not necessarily represent an ideal

medical technical description, and while they are included for completeness, they will be refined in future versions of the dataset.

B. Benchmark

During the model evaluation phase, experiments were conducted on a machine equipped with an NVIDIA A40 GPU. The selected models span multiple categories, including general-purpose multilingual models, domain-specific medical models, and models specifically trained for Brazilian Portuguese.

Large-Scale Models. To establish strong baselines, we evaluated proprietary models accessed via API, including GPT-4o, GPT-4o-mini, DeepSeek-672B [15], Sabia-3, and Sabiazinho-3 [16]. Due to their extensive training data and advanced architectures, these models serve as high-performance references. However, their use in the medical domain presents two significant challenges. First, privacy concerns arise since medical data often cannot be shared with third-party APIs. Second, the sheer volume of data used to train these models increases the risk of data leakage.

Generalist Multilingual Models. We considered InternLM3 [17], InternLM2 [17], Qwen [18], Gemma [19], and LLaMA 3.1 [20], which are designed for multilingual applications but lack explicit medical specialization. Their performance provides insight into the effectiveness of general-purpose large language models in medical reasoning.

Portuguese-Specific Models. We examined Sabia-7B [21], ChatBode-7B [22], Bode-7B [23], and CabraLLaMA¹, which are trained explicitly on Brazilian Portuguese data. This category assesses whether monolingual models outperform multilingual ones in medical tasks when language-specific training data is available.

Multilingual Medical-Specialized Models. We considered Apollo-7B [3], MMed-LLaMa-3-8B [4], MedLLaMA3², OpenBioLLM [24], and Slerp-Med³. These models incorporate medical training but are not specifically optimized for Portuguese. Their evaluation provides a comparative point for assessing the necessity of language adaptation in domain-specific settings.

Portuguese Medical-Specialized Models. DrBode-7B-360k and DrBode-7B-240k [7], as well as Clinical-BR-LLaMA [14], were included as the most relevant models trained with Brazilian Portuguese medical data. Their performance directly reflects the impact of adaptation within a low-resource linguistic context.

All models were evaluated using a temperature setting of 0.1 and a maximum token limit of 10, no quantization was performed for the open-source models. This selection enables a comprehensive analysis of how different model types—ranging from large closed-source systems to specialized Portuguese medical models—perform on the proposed benchmark.

The prompt provided to the models instructed them to receive a medical question and select the correct answer. It also included three examples demonstrating the expected response format, which consisted solely of the corresponding letter of the correct alternative, from the maximum number of options observed across all exams (A, B, C, D, or E). Additionally, the prompt specified that the model should refrain from providing justifications for its selection or including any text beyond the chosen alternative letter. The question statement was then inserted, and if the question contained an image, its description was appended immediately following the statement, with no further instructions. Lastly, the alternatives were presented, and each question was submitted to the model individually, adhering to the prescribed structure.

Upon obtaining the models' responses, the evaluation was conducted using solely the first letter generated by the model. Thus, even if multiple letters were produced, the answer was assessed based only on the initial letter provided. The primary metrics employed in this analysis included general accuracy, defined as the proportion of correctly answered questions relative to the total number of questions, accuracy based on image presence, and the relative contribution to accuracy based on image presence.

IV. RESULTS AND DISCUSSION

In this section, we present the results across different exam years, including overall accuracy based on the presence of images in the questions, in order to identify key patterns in the models' performance and offer valuable perspectives into the strengths and limitations of current language models in the medical domain. In the following subsections, we discuss the overall accuracy across categories, then explore the impact of visual elements in the exam questions, and provide an analysis of the question types that presented substantial challenges for the models.

A. General model accuracy

The individual performance of each model is presented in Table I and shows high variability in effectiveness across different categories. Models in the Large-Scale Models category, such as GPT-4o and DeepSeek, exhibit superior performance, with GPT-4o achieving an accuracy of 81.94%. These results are consistent with what is expected of such models since they benefit from extensive training on diverse datasets, which, although beneficial, raise a concern regarding data leakage of the dataset during training. Since these models are also often large-scale, they need to be accessed via third-party APIs, raising concerns regarding the confidentiality of medical data, which is often subjected to strict privacy regulations. These privacy concerns limit the practical applicability of these models in medical environments.

In contrast, models within the Generalist Multilingual Models category, such as Qwen 2.5 and InternLM3, show noticeably lower performance. Qwen 2.5 achieves an accuracy of 55.50%, which, although moderate, is a clear decline when compared to the performance of the large-scale models. The

¹<https://huggingface.co/botbot-ai/CabraLlama3-8b>

²<https://huggingface.co/johnsnowlabs/JSL-MedLlama-3-8B-v2.0>

³<https://huggingface.co/shanchen/llama3-8B-slerp-med-chinese>

TABLE I: General performance of various models across different categories, measured in terms of accuracy, precision, recall and f1 score. Entries marked as Failed to perform denote models that encountered issues, such as hallucinations and the inability to answer the benchmark questions.

Category	Model	Accuracy	Precision	Recall	F1
Large-Scale Models	GPT-4o	81.94%	82.34%	81.65%	81.99%
	DeepSeek	78.94%	77.71%	79.14%	78.42%
	Sabia 3	74.33%	73.62%	73.57%	73.60%
	Sabiazinho 3	72.02%	70.22%	70.17%	70.20%
	GPT-4o-mini	66.80%	69.23%	63.71%	66.35%
Generalist Multilingual Models	Qwen 2.5	55.50%	55.68%	54.19%	54.92%
	InternLM3	54.57%	53.57%	53.11%	53.34%
	InternLM2	39.74%	34.33%	33.49%	33.91%
	LLaMA 3.1	0.23%	16.67%	0.20%	0.39%
Portuguese-Specific Models	ChatBode-7B	41.51%	40.11%	40.00%	40.06%
	Bode 3.1	41.42%	54.39%	38.75%	45.24%
	Bode-7B	25.59%	35.35%	21.81%	26.97%
	Cabra LLaMa	37.28%	42.73%	40.28%	41.47%
	Sabia-7B	1.61%	24.46%	2.42%	4.40%
Multilingual Medical-Specialized Models	MMed-LLaMa-3-8B	33.97%	35.03%	30.89%	32.83%
	Apollo-7B	0.23%	12.38%	0.20%	0.39%
	MedLLaMA3	0.15%	20.00%	0.13%	0.27%
	LLaMa 3 SLERP	0.54%	12.00%	0.46%	0.89%
	OpenBioLLM	Failed to perform	Failed to perform	Failed to perform	Failed to perform
Portuguese Medical-Specialized Models	DrBode-7B-360k	41.66%	40.59%	36.97%	38.69%
	DrBode-7B-240k	41.51%	40.11%	40.00%	40.06%
	Clinical-BR-LLaMA	Failed to perform	Failed to perform	Failed to perform	Failed to perform

performance drop is even more pronounced in LLaMa 3.1, which only achieves an accuracy of 0.24%, by consistently picking the letter *E* as the final answer. This stark contrast highlights the challenges of applying generalist multilingual models to specialized tasks, particularly those that require domain-specific knowledge and nuance.

The differences in model performance can also be attributed to the number of parameters each model contains. For example, InternLM3, which has 8 billion parameters, achieves an accuracy of 54.57%, indicating that a larger model size can contribute to better performance on certain tasks. The larger number of parameters in GPT-4o likely enables the model to capture more complex patterns and nuances, making it particularly well-suited for specialized tasks, such as medical exams.

Models in the Portuguese-Specific and Portuguese Medical-Specialized Models categories exhibit similar trends, with performance metrics indicating that, while these models are adapted to understand the Portuguese language, they struggle with tasks that require specific medical knowledge or problem-solving skills. For example, DrBode-7B-360k and DrBode-7B-240k show reasonable accuracy values (around 41%), but they fail to demonstrate proficiency in task-specific knowledge by either failing to include medical data or failing to understand how to answer the question appropriately in the desired format. This is especially evident in the case of Clinical-BR-LLaMA, which is tuned with a large corpus of Brazilian Portuguese medical data but specifically for clinical note completion. This model excels in providing medically relevant information, yet it struggles with tasks requiring broader generalization..

Meanwhile, the Multilingual Medical-Specialized Models

category exhibits particularly low performance across all models, including MMed-LLaMa-3-8B and Apollo-7B, which achieve accuracies of 33.97% and 0.23%, respectively. Suggesting that, while the number of parameters or the presence of domain specific data can be considered a major component in the analysis of the difficulty of adapting general-purpose models to specialized domains, the quality of the training data and cultural nuances specific to Portuguese may also be important so that the model is capable of properly learning. OpenBioLLM, for instance, failed to perform due to significant issues with hallucinating answers and an inability to generate responses that adhered to the requested format.

B. Accuracy by exam year

Figure 1 presents a longitudinal analysis of language model performance on medical benchmarks based on Brazilian performance tests from 2011-2024. The first thing that can be noticed is that all considered models demonstrate a consistent performance decline on 2024 medical tests, suggesting these assessments may incorporate recent medical knowledge, terminology, or practices not fully represented in model training data. GPT-4o maintains the highest accuracy (around 78%) on 2024 tests while still experiencing a 10-percentage-point decline from its performance on 2022-2023 tests.

It is also interesting to notice that performance fluctuations across specific years (2012, 2016) are consistent across all models, indicating these variations likely reflect differences in test difficulty rather than model limitations. The significant performance gap between 2016 and 2021 tests suggests either a potential shift in test difficulty or the introduction of new medical concepts during this period. Furthermore, the rela-

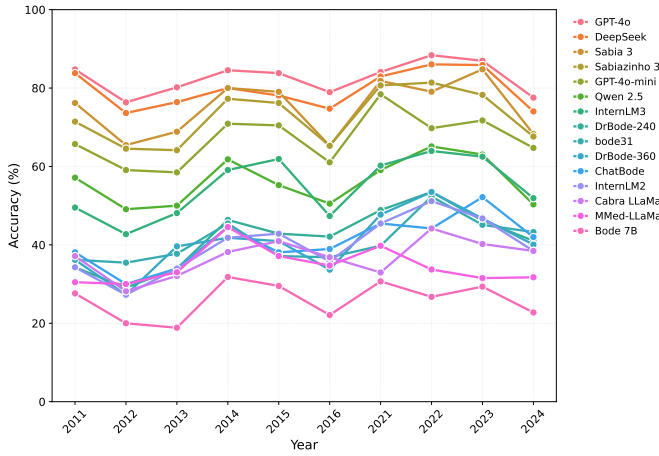


Fig. 1: Accuracy of language models evaluated across both exams in multiple years. Only models that achieved sufficient accuracy were included in the analysis.

tively stable performance observed in the 2014-2015 period, followed by a decline in 2016, suggests that certain medical concepts or question structures from these years may have been better represented in the models’ training data.

The stratification between model performance tiers remains consistent across test years. Frontier models (GPT-4o, DeepSeek) consistently outperform specialized medical models like MMed-LLaMa on Brazilian medical tests, raising questions about whether general knowledge breadth might sometimes be more valuable than domain specialization for medical reasoning, as it incorporates task-specific information into the model.

These patterns indicate that models trained primarily on more recent datasets may face challenges with older medical terminology and struggle with structural variations in exam formats. As such, models must be exposed to various question structures and content types, including variations in how answer choices are presented, a case that often happens in large-scale models.

C. Accuracy by image presence

Table II presents a comparative analysis of model accuracy with and without images on the compared models and shows a notable decrease in accuracy when medical images are incorporated into the questions, with a loss ranging from 5.57% to 19.88% of accuracy across most models.

This pattern suggests that models without visual capabilities face difficulties in understanding the complexities of medical images when presented with only textual descriptions. This raises an important question: How would these models perform if visual language models (VLMs) were employed? It is also crucial to acknowledge that the image descriptions provided to the models were not optimized for medical accuracy and were not written by medical professionals, which likely contributed to the observed performance decline.

TABLE II: Model accuracy on questions requiring image data for the evaluation.

Model	Without Images	With Images	Difference
GPT-4o	83.73%	64.46%	-19.27%
DeepSeek	80.09%	67.77%	-12.32%
Sabia 3	76.02%	57.85%	-18.17%
Sabiazinho 3	73.56%	57.03%	-16.54%
GPT-4o-mini	68.64%	48.76%	-19.88%
Qwen 2.5	56.70%	43.80%	-12.89%
InternLM3	55.42%	46.28%	-9.14%
DrBode-240	42.20%	36.36%	-5.84%
ChatBode	42.20%	34.71%	-7.49%
DrBode-360	41.27%	35.54%	-5.73%
Bode 3.1	42.11%	34.71%	-7.41%
InternLM2	40.34%	33.88%	-6.46%
Cabra LLaMa	37.80%	32.23%	-5.57%
MMed-LLaMa	34.49%	28.93%	-5.57%
Bode 7B	25.85%	23.14%	-2.71%
Sabia 7B	1.54%	2.47%	0.95%
LLaMa 3 SLERP	0.51%	0.82%	0.31%
Apollo	0.25%	0.00%	-0.25%
LLaMa 3.1	0.25%	0.00%	-0.25%
MedLLaMa3	0.16%	0.00%	-0.16%
OpenBioLLM	0.00%	0.00%	0.00%
Clinical-BR-LLaMA	0.00%	0.00%	0.00%

A particularly noteworthy observation is that despite the limited visual data and the non-clinical nature of the image descriptions, models such as DeepSeek still achieved 67.77% accuracy when tasked with processing images. This raises concerns about potential data leakage, as the model may have been inadvertently seen the questions in its training data.

These findings have important implications for clinical deployment in Brazilian healthcare settings, where the accurate interpretation of diagnostic images is crucial. The considerable performance gap between text-based and image-based reasoning suggests that even advanced models require substantial improvements in multimodal processing capabilities before they can be reliably used in image-intensive medical fields such as radiology or dermatology.

To address these challenges, we plan to enhance our benchmark in the future by incorporating actual image data. This will allow for a more comprehensive evaluation of VLMs in medical reasoning tasks and provide insights into their strengths and limitations when handling multimodal input.

D. Question Types Posing Significant Challenges to Model Performance

To identify the most challenging question types for the evaluated language models, the medical field categorized questions incorrectly answered by all models using GPT-4o mini. This analysis reveals distinct patterns in model limitations across medical specialties. Pediatrics emerges as the primary stumbling block with ten questions failed by all models, followed by gynecology and obstetrics (five questions each) and collective health (four questions).

The predominance of pediatric failures is likely caused by the field’s unique combination of developmental physiology, age-specific presentation variations, and non-verbal assessment requirements. In addition, Brazilian pediatric protocols

may further diverge from international standards represented in training data, particularly regarding growth metrics and vaccination schedules. On the other hand, the issue with collective health questions exposes fundamental weaknesses in contextual reasoning about Brazilian healthcare infrastructure, resource allocation, and population-level interventions.

Gynecology and Obstetrics are other areas of concern, with both appearing five times. These fields deal with a variety of conditions related to female reproductive health, which often involve sensitive and complex medical topics. Questions in these areas could include anything from hormonal disorders to complicated childbirth procedures, requiring the models to distinguish subtle differences in terminology and clinical presentation.

The frequency with which these categories appeared in the dataset, combined with the depth of medical knowledge required, indicates that language models might still struggle in fields that demand a high degree of specialization and context-sensitive understanding.

V. CONCLUSION AND FUTURE WORKS

This work presents a novel benchmark for evaluating medical language models in Brazilian Portuguese, addressing a critical gap in the assessment of AI systems for healthcare applications in non-English contexts and revealing several important insights that have significant implications for the development and deployment of medical AI in Brazil and other Portuguese-speaking regions. All models had substantial accuracy reductions when handling medical images, suggesting a step forward in the analysis of VLM. In addition, an analysis of model performance across different medical specialties identified pediatrics, gynecology, obstetrics, and collective health as particularly challenging domains, showcasing that current models struggle with Brazil-specific clinical practices and socioeconomic healthcare determinants.

In future work, we intend to expand this benchmark to encompass additional medical specialties and conduct a vision-language model-based analysis to better understand multimodal reasoning capabilities in medical contexts.

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